

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	01/10.2026	Duration:	60 minutes

Code of Conduct

I Divya Dharshini V S (192511308) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

AI-Based Network Intrusion Detection and Threat Monitoring System

1. Identify the Project Vision and Stakeholders

1.1 Define the project's vision, objectives, and identify all key stakeholders involved.

The project's vision is to design an AI-Based Network Intrusion Detection and Threat Monitoring System. The system's main objective is to monitor or detect undesirable or malicious activities that take place in a network. The software will target and check the network traffic in a system and, at the same

time, raise an alarm for any suspicious activities. It will also help the organization protect its network against attacks and secure the company's data.

My project objectives include:

- Monitoring the network traffic on a regular basis.
- Looking out for unusual activities in the network.
- Identification of common attacks like port scans and unauthorized access.
- Monitoring of intrusive network traffic.
- Using artificial intelligence to detect possible network attacks.
- Raising an alarm with the general description of the possible attack.
- Providing general information on the detected attack to help the network security personnel respond fast and appropriately.
- Reducing the burden of manual traffic inspection to help enhance the performance of the network.
- Storing the detected attack data for later inspection.

The stakeholders of my project include:

- System Administrator: They monitor the system and set up the network.
- Security Analyst: They monitor the alerts and respond to detected attacks.
- Network Administrator: They manage and monitor the network.
- Management: They ensure the network systems are protected.
- End users: They use the network systems and services provided.
- Development Team: They develop and maintain the system.
- IT Security Team: They handle and deal with security issues in the network.
- System Owner: They monitor the requirements and ensure that the system meets the performance standards.

The primary stakeholder involved in the project is to develop a system that satisfies the most required stakeholders. For instance, the system administrator and the security analyst can use the intrusion detection to monitor and respond to possible attacks in the network. Therefore, the project's objectives focus mainly on detecting and monitoring network attacks.

2. Create Epics and User Stories

2.1 Break down the requirements into Epics and formulate well-structured User Stories using the format:

As a <user>, I want <feature>, so that <benefit>.

For my project, I divided the requirements into several Epics to simplify the development process. I tried to make the stories short and simple, focusing on the actual actions that would be performed by users of my network monitoring system. In addition, this approach will help me prioritize the Epics during the development of the system. The main Epics and their associated User Stories are as follows:

a) Epic 1 – Network Monitoring

- As a Network Administrator, I want to monitor network traffic, so that I can know what activities are happening in the network.
- As a System Administrator, I want to view network activity details, so that I can check whether the network is working normally.

b) Epic 2 – Threat Detection

- As a Security Analyst, I want the system to detect suspicious network activities, so that I can identify possible attacks.
- As a Security Analyst, I want the system to identify different types of threats, so that I can understand the type of attack.
- As a System Administrator, I want the system to detect unusual traffic patterns, so that unknown or changed attacks can also be noticed.

c) Epic 3 – Alert Management

- As a Security Analyst, I want to receive alerts when a threat is detected, so that I can take action quickly.
- As a System Administrator, I want to see the details of an alert, so that I can understand what caused the alert.
- As a Security Analyst, I want to mark an alert as checked or resolved, so that I can keep track of handled threats.

d) Epic 4 – Threat Reports

- As a Security Analyst, I want to view previous threat records, so that I can check past security incidents.

- As a Management User, I want to see a summary of detected threats, so that I can understand the security condition of the network.
- As a Security Analyst, I want to export threat information, so that I can use it for further analysis or reporting.

3. Define Acceptance Criteria

3.1 Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

The acceptance criteria for the selected User Stories are as follows:

- a) User Story: As a Network Administrator, I would like to monitor the network traffic to know what is happening in the network.
 - Given the network is connected, when the administrator accesses the monitoring system, it shall display the current network traffic.
 - The network traffic shall include the relevant traffic information.
- b) User Story: As a Security Analyst, I would like to have suspicious network activities detected to know what is happening in the network.
 - Given the network traffic is monitored, when suspicious network activity is detected, the system shall identify it as a potential threat.
 - The identified network security threat shall have basic information about it displayed.
- c) User Story: As a Security Analyst, I would like to be alerted about the detected threats so that I can respond to them in time.
 - Given the system has detected a network security threat, when the system finishes the detection procedure, it shall alert the Security Analyst.
 - The alert message shall contain all the relevant information about the detected threat.
- d) User Story: As a System Administrator, I would like to be able to mark alerts as acknowledged or resolved to have a reference of what has been dealt with.
 - Given the system has generated an alert message, when the system administrator acknowledges or resolves the alert, its status shall be changed accordingly.
 - The updated information about the alert shall be accessible and visible to the system administrator.

The given acceptance criteria help me ensure that the most critical aspects of the designed system will be functioning correctly. They also contribute to the project by making the User Stories testable due to the clear description of what actions should be performed to verify that a feature is implemented correctly and works as expected. If a feature passes all the given acceptance criteria, then it can be considered as complete.

4. Prioritize the Product Backlog

4.1 Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

In my project, I would prioritize the Product Backlog according to the features' importance for the system. I will prioritize developing the set of functions necessary for detecting and reporting threats over the ones providing means for displaying historical information. Moreover, I will prioritize those features which do not depend on the implementation of other functions.

Product Backlog Prioritization:

a) Network Traffic Monitoring – Priority 1

- This feature is the basis for the whole system since it enables the collection of all network traffic data necessary for detecting threats. Therefore, the system should have the capability of displaying the relevant network data on its interface.

b) Threat Detection – Priority 2

- This system capability is critical for the project since it enables detecting the network traffic anomalies that may indicate the occurrence of a network attack. The system's feature of "Network Traffic Monitoring" should be implemented before proceeding to this function.

c) Threat Alert Generation – Priority 3

- This feature is also critical since it allows informing the system's users about the detected irregularities. Thus, the system should have the capability of notifying the security service about the potential network threats.

d) Threat Details and Analysis – Priority 4

- This system function allows the network security analysts to learn the details of the detected anomalies. It is necessary for them to have this information to understand the characteristics of the irregularities that have been detected by the system.

e) Threat History – Priority 5

- This feature is valuable to network security analysts since it allows them to review the history of network threats and see the details of each particular threat. This way, the system offers an opportunity to analyze the characteristics of threats and possible ways of their appearance.

5. Plan the Sprint Backlog

5.1 Select high-priority user stories for the first sprint and divide them into actionable development tasks.

For my first sprint, I will prioritize the User Stories network traffic monitoring, suspicious activity detection, and threat identification. Since they all are closely related and can provide the basic monitoring functionality, I will develop them first. They will allow me to get the fundamental version of the software under my wing. The remaining User Stories will be taken up in the following sprints.

The User Stories for the first sprint are:

a) Network Traffic Monitoring

- Create the network monitoring segment.
- Collect the required traffic data.
- Display the obtained traffic data in the system.
- Verify the correctness of collected data.

b) Suspicious Activity Detection

- Prepare the data for detecting threats.
- Find normal and suspicious network activity.
- Apply the AI-based threat detection method.
- Test the system with various activities.
- Check if suspicious activities are correctly identified.

c) Threat Identification

- Identify the detected threat type.
- Display the basic information about the threat.
- Verify the detection with test data.
- Ensure results are easy to comprehend.

d) Threat Alert Generation

- Create the alert feature.
- Trigger alerts upon threat detection.
- Show the information about the detected threat.
- Verify if alert generation works correctly.

e) Basic Threat Record

- Store the data about detected threats.
- Display the previously stored threat data.
- Verify if the storing mechanism works correctly.

The primary goal of the first sprint is to create the basic monitoring and detection functionalities. The development tasks were formulated in such a way that multiple smaller tests can be made to verify each individual step. After developing these functions, I should get a simple program that is able to monitor the network traffic, detect possible threats, and alert the user.

6. Estimate Effort Using Story Points

6.1 Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

I would use the Fibonacci sequence for estimating the effort required to develop each User Story. The Fibonacci sequence that I will use for estimation includes values of 1, 2, 3, 5 and 8 where higher values indicate higher effort required to finish a User Story. It will help me to get an idea of the relative amount of work required to develop each feature prior to commencing the first sprint.

The story points for each of the selected User Stories are listed below:

a) Network Traffic Monitoring – 3 points

- Collecting the basic information about the network traffic will require moderate effort.
- Some testing will be needed to ensure that the network traffic is being collected appropriately.
- Therefore, I have assigned 3 points to this user story since it will require moderate effort.

b) Suspicious Activity Detection – 8 points

- Activity detection is one of the main features of my project but it is also the most complex feature.

- This will include AI-based detection process that requires extensive testing.
- Thus, I have assigned 8 points to this User Story since it will require the highest effort to develop.

c) Threat Identification – 5 points

- Following the detection of suspicious activities, the system will have to identify the possible threats. The identification results will have to be presented clearly to the security analyst.
- This will include testing various categories of suspicious activities to ensure that the results are consistent.
- Therefore, I have assigned 5 points to this User Story.

d) Threat Alert Generation – 3 points

- A threat detection alert will have to be launched whenever possible threats have been identified.
- I will have to ensure that the alerts display simple information such as the type of the threat detected. Thus, I have assigned 3 points to this User Story since it will require moderate effort.

e) Basic Threat Record – 3 points

- The basic information regarding detected threats will have to be recorded for future reference.
- Once again, this feature will require extensive testing to ensure that the recorded information is easily accessible. Thus, I have assigned 3 points to this User Story since it will require moderate effort to develop.

Based on the story points listed above, the cumulative effort required to develop these User Stories is 22 points. Suspicious activity detection has been assigned the highest story points since it involves AI-based detection and extensive testing. It will assist me in ensuring that the workload is well distributed.

7. Present the Sprint Goal and Expected Deliverables

7.1 Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

For my first sprint, I would like to focus on developing the basic working part of my AI-Based Network Intrusion Detection and Threat Monitoring System. I want to achieve the creation of a system that can monitor the network traffic, recognize possible suspicious activities, and notify about potential threats. At the end of the first sprint, these features should be functional and ready for the testing phase.

My sprint goal is to

- Monitor the network traffic and collect the required data,
- Identify suspicious activities,
- Utilize the planned AI-based strategy to detect potential network threats,
- Recognize the possible type of identified threats,
- Generate an alert regarding the detected suspicious activities,
- Display the essential information about the potential threat,
- store the basic data about the detected threats for further analysis,
- Thus, the first sprint will result in several features that I will code and test.

The expected deliverables of the first sprint include:

- Working Network Monitoring Feature – the software will be able to collect and display the primary data about the network traffic;
- Threat Detection Feature – the system will detect and highlight any suspicious activities;
- Threat Identification Feature – the software will indicate the possible type of the detected threats;
- Alert Feature – the system will generate an appropriate alert message;
- Threat Record Feature – the system will store the basic information about the detected threats;
- Basic Testing Results – the features developed in the first sprint will undergo the testing stage to ensure their proper functioning;
- Working Sprint Version – the working version of the first sprint will be prepared for the demonstration at the end of the first sprint.

I believe that at the end of the first sprint, I will be able to create a working version of my system that will be focused on network monitoring and detecting potential threats. In the next stages, the system will include more functions regarding the detection of network threats and analyzing the collected data in more detail and in greater depth.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25